Ali NCIBI

Ph.D.
Personal homepage
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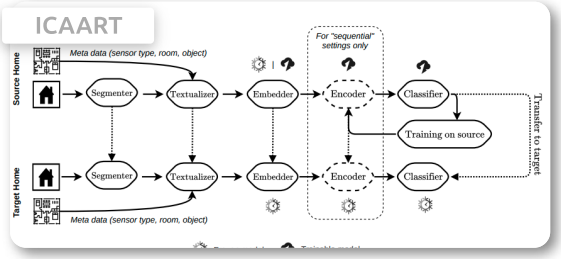
About Me

Researcher and engineer interested in machine learning, data, and intelligent systems. I enjoy building systems, exploring ideas, and turning complex problems into practical solutions.

Research Interests

- Machine Learning
- Human Activity Recognition
- Intelligent Systems
- Applied AI for real-world environments

Selected publications

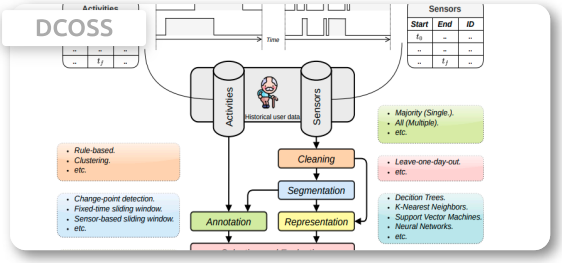


Towards Domain-Robust Activity Recognition using Textual Representations of Binary Sensor Events

Ali Ncibi

18th International Conference on Agents and Artificial Intelligence ***(ICAART)****, 2026.*

PDF Code BibTex

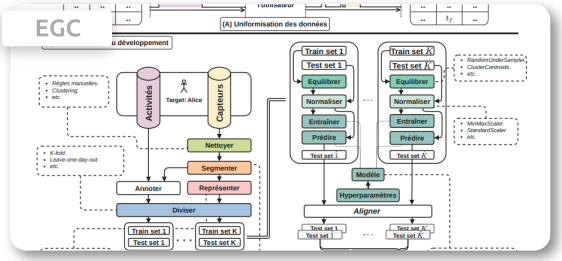


FlowAR: A Framework for Data-Driven Development of Human Activity Recognition Systems using Binary Sensors

Ali Ncibi, Luc Bouganim, Philippe Pucheral

*21st International Conference on Distributed Computing in Smart Systems and the Internet of Things **(DCOSS-IoT)**, 2025.*

PDF BibTex



FlowAR: une plateforme uniformisée pour la reconnaissance des activités humaines à partir de capteurs binaires

Ali Ncibi, Luc Bouganim, Philippe Pucheral

*25ème conférence francophone Extraction et Gestion des Connaissances **(EGC)**, 2025.*

PDF BibTex

Professional & Personal Projects

Flux Media Player

Personal online project — 12/2025 –Ongoing

A media player and library organizer for personal streaming servers, enabling users to stream and manage IPTV content (Xtream and M3U) through a beautiful, responsive, and unified interface.

Tech: Node.js, Nuxt, Vercel

Live Demo

Massirah CV Builder

Personal online project — 06/2025 –Ongoing

A beautiful, responsive, multilingual web application that helps users create clean, ATS-optimized resumes, enhanced with AI-powered job matching and smart suggestions to improve and tailor CVs for specific roles.

Tech: Node.js, Express, PostgreSQL (via Supabase), React, Heroku

Live Demo

FlowAR (Flows of Activity Recognition)

Part of PhD thesis, INRIA / Université Paris-Saclay — 03/2023 –02/2026

A modular, data-driven experimental framework designed to support reproducibility and scientific rigor in the development and evaluation of activity recognition systems for PhD research.

Tech: Scikit-learn, scikit-optimize, PyTorch Lightning, SciPy, NumPy, Pandas, Matplotlib, Plotly, Streamlit

Task-oriented dialog management system (chatbot)

Freelance mission as a data science engineer, SouthPigalle (later part of Metyis Group), Paris. — 10/2021 –05/2022

Designed, implemented, and validated a task-oriented chatbot prototype for handling user queries within a domain-specific ontology (case study: Van Cleef & Arpels jewelry).

Tech: PyTorch, Streamlit

Temporal-links extraction between medical concepts from French medical text

End-of-study internship, INSERM, Paris. — 03/2021 –08/2021

Designed, implemented, and evaluated a deep learning-based approach for temporal relation (T-link) extraction from French clinical text corpora.

Tech: PyTorch

Style transfer between music accompaniments in MIDI

Research internship, LTCI Lab, Télécom Paris. — 09/2020 –02/2021

Applied deep learning techniques to automatic music generation. Tweaked a generative model (Groove2Groove: https://groove2groove.telecom-paris.fr) for the task.

Tech: TensorFlow

CNN-based registration of 2D brain tumor images

Academic project, LTCI Lab, Télécom Paris. — 01/2020 –03/2020

Studied a CNN-based approach for brain tumor image registration. Evaluated different training losses and explored the resulting transformations on examples from the BraTS MICCAI brain tumor dataset.

Tech: TensorFlow

Arrhythmia detection from ECG data

Academic project, ENIT, Tunisia. — 02/2019 –04/2019

Designed and implemented a neural network for arrhythmia detection through the analysis of patient ECG recordings. Evaluated performance on examples from the MIT-BIH Arrhythmia Database.

Tech: TensorFlow

Network standards for IoT applications in healthcare

Academic project, ENIT, Tunisia. — 03/2018 –05/2018

Reviewed and discussed networking trends for IoT in healthcare applications.

Tech: Networking / IoT

Skills

Data Science Toolkit

- Supervised and unsupervised machine learning (classification and clustering)
- Descriptive statistics and statistical hypothesis testing
- Experience working with multiple data modalities: tabular, image, text, and audio
- Experience with Scikit-learn, PyTorch, and Python data science libraries; model training and deployment

Programming Languages

- Python
- JavaScript
- C/C++, C#
- SQL

Web & Mobile Development

- HTML5 and CSS3
- React
- Node.js
- Vue.js (Nuxt.js)
- Ionic / Capacitor

Tools

- Version control: Git
- Containerization: Docker
- Operating systems: Linux, macOS, Windows
- Cloud platforms: AWS, Google Cloud Platform (GCP), Heroku, DigitalOcean
- Productivity & authoring tools: Microsoft Office, Adobe Creative Suite, LaTeX

Education

PhD in Computer Science

Univeristy of Paris Saclay (UPS) — UVSQ/INRIA, Versailles/Saclay, France

2023 - 2026

Focus on the design of structured data-driven pipelines and the evaluation of semantic representation strategies for cross-home human activity recognition from IoT traces

MSc in Data Science

Telecom Paris, Institut Polytechnique de Paris (TPT/IPP) — Paris, France

2019 - 2022

Focus on numerical optimization, statistical machine learning, computer vision, natural language processing, graph mining, programming, and social & economic sciences.

MSc in Information Technology and Telecommunication

The National Engineering School of Tunis (ENIT) — Manar, Tunisia

2017 - 2019

Focus on signal transmission and processing, networking techniques, programming, databases, and system administration.

Preparatory class for the tunisian/french Grandes Ecoles systems

Institut préparatoire aux études scientifiques et techniques (IPEST) — Marsa, Tunisia

2014 - 2016

Specialized in mathematics and physics.